Task	Depth / Width / Hidden Width	Epochs	Augmentation	Training Time	Baseline Test Accuracy	Final Test Accuracy
MNIST	10 blocks / 128 / 256	10	None	1.5 min	95.06%	98.35%
CIFAR-10	10 blocks / 128 / 256	30	Random horizontal flips and padded random crops	2.1 min	48.61%	58.52%
ImageNet32	3210 blocks / 128 / 256	15	Random horizontal flips and padded random crops	20.9 min	6.77%	11.01%

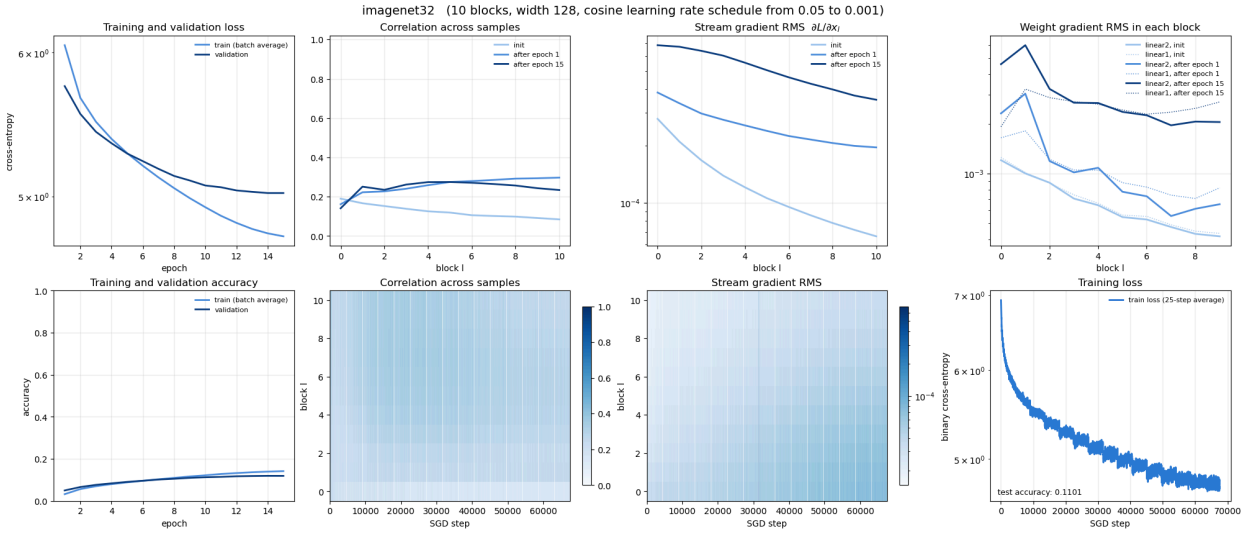
Components	Final Settings
Optimizer	AdamW, learning rate 0.001, batch size 256 .
Schedule	Cosine annealing, stepped each epoch, with minimum learning rate 0.00001.
Loss	Cross-entropy.
Regularization	Weight decay 0.01.
Input Normalization	Pixel values scaled from [0, 255] to [−1, 1].
Activation	SwiGLU.
Initialization	Xavier-normal linear weights and zero biases.
Color-Image Augmentation	Horizontal flip with probability 0.5; four-pixel padding followed by a random 32×32 crop.

2. What Mattered Most

a. Adding a scheduler and optimizer was definitely worth it. My training loss between change 4 and 5 started to plummet more. Log 4 and 5.

b. Adding image augmentation. It helps the network to be less overfit. Log 6 and 7.
c. Changing the activation function. I made more of an expressive change than tanh. Log 1, 2, and 3.

3. Training Dynamics



4. What Did Not Work

a. Changes in some variables like depth and and width. They seemed to be great the entire time based on each graph before the start.
b. Having more complicated activation functions on the first iterations. It changed from ReLU to SiLU to SwiGLU and the accuracy got worse.
c. Ignoring the obvious overfitting problem for way too long. They needed to be taken care of faster despite the more complicated features that were added.

5. Across the Three Tasks

a. MNIST: The dataset is so simple and small that it did not even need much to be really good. I simply stopped working on it XD
b. CIFAR-10: The dataset was harder to work on due to its bigger size and complexity. I still was able to change the accuracy substantially but there was a point where good methods learned in lectures were adding little change. Even if I missed some methods.
c. ImageNet 32×32: SO MANY IMAGES. Whenever a breakthrough was made, the results did not feature much of a change. More complication model feature are definitely needed for this model to be at less 50% accurate
d. What's Told: They are better for simple things. Especially for flatten images. The more complicated the dataset got, the worse the model expressed in results. I tested a Patch-Based MLP in Change 8. This method expressed an option resembling

convolutional modeling. It shows exponentially better results, even when not fully run though outside issues and time. Image inputs should be changed more for better generalization and point of view in the model.

6. Explanations of Anything Beyond the Basics

a. AdamW and Cosine Annealing Learning Rate are explained in log 5.
b. Image flipping and image pad-cropping is explained in log 6 and 7.
c. Un-recorded: Patch-Based MLP is explained in log 8.

7. Note To Professor

a. Awesome project. This project helped put together and reinforce a lot of the ideas from lectures.
b. The leader board is definitely motivating.